Dissertation Type: research

DEPARTMENT OF ENGINEERING MATHEMATICS

COVID-19 Fake News and Health Misinformation Detection on Twitter

Ali Almalki

Supervised by: Dr. Ryan McConville

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science in the Faculty of Engineering.

Monday 4th October, 2021

Contents

1 Contextual Background **1**
1.1 Introduction . . . 1
1.2 Problem Statement . . . 1
1.3 Motivation . . . 2
1.4 Methodology . . . 3
1.5 Objectives . . . 3
1.6 Challenges . . . 4
1.7 Summary . . . 4

2 Technical Background **5**
2.1 Machine Learning and NLP Background . . . 5
2.2 Word Embeddings Algorithms . . . 7
2.3 Graph Terminology and Representation . . . 8
2.4 Representation Learning on Graphs . . . 9
2.5 Supervised Machine Learning Classifications Algorithms . . . 10
2.6 Fake News Detection Relevant Literature . . . 13
2.7 Dataset . . . 15

3 Project Implementation **17**
3.1 Introduction . . . 17
3.2 Project Management . . . 17
3.3 Dataset Creation . . . 18
3.4 Dataset Overview . . . 19
3.5 Architecture of the Experimental Setup . . . 22
3.6 Text Modelling . . . 25
3.7 Graph Modelling . . . 26
3.8 Evaluation Methods . . . 28

4 Project Results and Critical Evaluation **31**
4.1 Introduction . . . 31
4.2 Approach to Result Generation . . . 31
4.3 Baseline Model Results . . . 31
4.4 Removing Hashtag Keywords Results . . . 32
4.5 ML Models with Independent Features Results . . . 32
4.6 ML Models with Combined Features Results . . . 34
4.7 ML Models with Node2Vec Feature Representation Results . . . 36
4.8 Results Analysis . . . 36
4.9 Performance summary and conclusion . . . 37
4.10 Results Evaluation . . . 37
4.11 Assessment with regards to project objectives . . . 39

5 Conclusion **41**
5.1 Summary . . . 42
5.2 Project Status . . . 42
5.3 Further Work . . . 43

List of Figures

1.1 A tweet portraying false information that can mislead and manipulate many individuals who believe this statement is true [20]. 3

2.1 A figure illustrating node embedding learning methodology where we have the encoder (ENC) that maps nodes to a low-dimensional embedding space in a way that show the relative positions of the nodes in the original graph. [17] 9

3.1 A screenshot showing the repository status on my local machine 17
3.2 A screenshot showing the repository log on my local machine 18
3.3 The first 10 rows of the dataset 19
3.4 Nested JSON Dataset Structure 19
3.5 A bar chart showing label count 20
3.6 A sample of the tweet text 21
3.7 Tweet text length for each label 21
3.8 Most frequent words in the dataset 22
3.9 The architecture of the proposed pipeline. 23
3.10 Important features selection. 25
3.11 A graph representing the tweet ID and its retweets/replies connections 26

4.1 Usage of Word2Vec features generates the most accurate results 33
4.2 Comparison of graph-based and text-based modelling 38

List of Tables

3.1 The classifiers with various independent feature representation 26
3.2 Confusion matrix. 28

4.1 DT Classifier - Baseline Model 31
4.2 After removing hashtag keywords - DT Classifier 32
4.3 BoW-based feature with various classifiers 32
4.4 TF-IDF based feature with various classifiers 33
4.5 Word2Vec-based feature with various classifiers 33
4.6 Combined features with SVM classifier 34
4.7 Combined features with MNB classifier 34
4.8 Combined features with GNB classifier 35
4.9 Combined features with DT classifier 35
4.10 Combined features with KNN classifier 36
4.11 Node2Vec-based feature with various classifiers 36
4.12 Average performance for all classifiers using independent features 38
4.13 Average performance for all classifiers using a combination of features 38

Executive Summary

During the COVID-19 pandemic, social networks have seen a substantial amount of false news [28]. Furthermore, people have been discussing numerous false remedies to cure COVID-19. However, these remedies are extremely dangerous to human health. The director-general of the World Health Organization calls it “infodemic" [31] because of the amount of misinformation, disinformation, and "false news" relating to COVID-19. With the enormous number of news regarding COVID-19 on the Internet, it is difficult for many to assess truthfulness. Moreover, the riots and panic shopping also occurred due to the propagation of “false news" [31]. In this thesis project, I aim to build an automated COVID-19 misinformation detection system and investigate the value of a social network structure compared to the text-based classification approach. I have implemented a variety of techniques to detect fake news and misinformation in tweets related to COVID-19. The research objective is to classify each tweet as either true/fake with various text feature representations techniques and graph structure to compare and evaluate their performances. The project is comprised of two parts which are text-based and graph structure-based fake news detection techniques. For the first part, I conduct five different classification algorithms relying on various embeddings techniques including BoW, TF-IDF and Word2Vec embeddings. For the second part, I represent the data in a graph structure and learn the feature representations for the nodes using the Node2Vec embedding algorithm, which can then be used for the downstream classification task. Different studies [29, 32] revealed that N-grams based features are efficient in identifying false information. The author in [33] uses TF-IDF feature to classify hoax news with 84.67% accuracy. The classifiers used in this paper are more modern KNN, MNB, LSVM, and logistic classification. In this thesis, it was concluded that the use of a combination of features in the GNB classifier has the most accurate performance. Moreover, text-based modelling outperforms graph-based modelling in terms of ROC score, accuracy as well as weighted average F1 score.
The overall aim was to produce a system that can auto-detect fake news in tweets related to COVID-19, improving the results obtained by previous studies.

The main achievements of the project are as follows:

- I learnt how to clean and prepare large scale data in JSON format and convert it into a suitable format to be ready for the study analysis.
- I wrote a total of 18898 lines of source code (in Python), implementing all the experiments analysed in this thesis.
- I learnt how to build an NLP pipeline including word tokenisation and removing stopwords from the tweet text.
- I wrote functions in python code that encode different text feature representation including BoW, TF-IDF and Word2Vec.
- I investigated and trained many classification models with different feature representation combinations.
- I represented the data as a network structure (graph) and modelled it to be used in the context of detecting false information.
- I applied continuous feature representations for the nodes in a graph to be used for the classification task.

Chapter 1

Contextual Background

1.1 Introduction

The pandemic has been accompanied by a massive wave of false news that made it more difficult for the public to comprehend the truth. People use different social networks to share information and communicate with each other. However, this also opens a room to propagate fake information on these platforms. The spread of fake news can be extremely dangerous, especially when facing the COVID-19 epidemic. Moreover, fake news about fake remedies to cure COVID-19 can be hazardous to human health and lead to human death [43]. Processing an enormous amount of text manually on these social media platforms, particularly Twitter, is troublesome. Therefore, it is essential to present automatic methods to analyze text and annotate if the news is fake or real. Fake or inaccurate information can also put a human body at risk of exposure to viruses and also cause anxiety, mental stress [40]. Besides, the purpose of spreading fake news can also be to mislead people to gain attention. Believing wrong information can be an easy task; however, refuting fake news is a challenging task – not many people ask for evidence to back up the statement.

In the digital era, the general public can be easily misled by fake news. This can be due to various reasons, such as lack of education and lack of interest in news verification. However, the propagation of fake news can be dangerous. Misinformation on COVID-19 can encourage individuals not to wear masks, take certain precautions, or even get a vaccine when these are the factors that can keep them safe. Public health messages should be considered since fake information can be posted as very concise headlines on the internet, especially on Twitter. Therefore, it is crucial to automatically differentiate real and fake news related to COVID-19 to avoid hazardous consequences to public health.

A recent study [39] reported that 1500 tweets were posted on Twitter regarding COVID-19, including 1274 were completely false and 226 were partially false. This implies that it may be unknown to the individuals that they are spreading fake news. If they knew that the shared information was fake, then it is probable that they would have preferred sharing true news. Hence, It is beneficial to understand the essence of propagating misinformation phenomenon on these social media platforms, especially Twitter. Twitter allows its users to share ideas easily, and posting a tweet or re-posting a particular tweet on Twitter does not require much effort. Nevertheless, words can be misused, and stories can be made up to make everyone believe using conspiracy theories. Research [34] shows that the number of conspiracy theories has increased due to the spread of COVID-19 worldwide. Another study [2] linked 5G technology with COVID-19, this has become dangerous since several people have started to burn the 5G towers in the United Kingdom. Finally, another study revealed that fake news on Twitter gets more response and spread faster than real news [2].

1.2 Problem Statement

The COVID-19 pandemic has caused many restrictions and induced different issues that impact people's social and economic lives. Employees have stopped in office work due to the fear of COVID-19. The root of this issue has become even more profound due to social media amplifying fake news. Moreover, the COVID-19 pandemic has become one of the significant events in all of our lives, and the impact has also been enormous due to approximately 199,376,990 million people being infected from the COVID-19 [1].

Rumours, false news, false data, lies, and gossips related to COVID-19 can provoke the public's actions because it is harming immune systems and physical health. This encourages people to believe in the conspiracies they see or hear on Twitter. Social media has become a medium that could negatively affect the public's mental health as well. For example, a recent study [4] demonstrated that 800 people have lost their lives globally, and 5,800 have been admitted to hospitals due to the fake news about COVID-19. Many individuals believed that the cure for the COVID-19 is by drinking methanol or alcohol-based cleaning products, not knowing how harmful these substances are to the body.

To provide reliable information, different research-led websites can make a big impact. For instance, more than 81 million articles are linked to the website of the Johns Hopkins Coronavirus Resource Center on Google. A research of 1000 participants about fake reports carried out in 2020 reveals that individuals prefer to readily spread news without being certain that it is genuine or false [25]. If the similar partakers realize which news is false. However, they are more likely than fake to share (genuine) news. This shows that false news is very desirable so that credible information may be distributed only.

COVID-19 has been a serious concern, and this gives some people an opportunity to introduce different kinds of fake news on social network platforms like Twitter. Some of the theories presented that the Chinese deliberately made COVID-19 to start a war against the USA. However, as time passed, many people beginning believing the whole idea of COVID-19 is not true, and it is merely a hoax by the governments – this idea has been exaggerated to create a sense of fear in people's minds [10]. Various techniques [3] have been proposed to identify fake news related to COVID-19 on Twitter automatically. This is a necessary step to stop or mitigate the storm of fake news on social media platforms. Nonetheless, the research on this topic is still evolving. Hence, it is beneficial to propose and study a new approach where I use both text-based and graph-structure based techniques that can detect false news on tweets regarding COVID-19. Both the content of the tweet (text) and the spreading of the tweet (the graph) would be useful indicators to detects fake tweets. The text contains keywords or phrases that indicate false information whereas the graph network structure could reveal how fast the tweet spreads as well as the most influential tweets. This can be identified by how many retweets/replies a tweet receives.

1.3 Motivation

The rationale behind this thesis study was derived from the considerable damage induced by the circulation of fake news concerning COVID-19 on social media platforms, primarily Twitter, and its impact on society's awareness. The COVID-19 pandemic has caused havoc to several individuals' lives, negatively impacting health and mental state due to the fake news about COVID-19. The propagation of fake news about the epidemic on Twitter, which has hundreds of users, can be devastating as many people are unsure what to believe. Consequently, this study is designed to detect fake news on Twitter related to COVID-19.

This research aims to build a robust automated misinformation detection system that identifies fake tweets related to COVID-19. This is highly crucial because the general public should verify the truthfulness of a tweet's claims before sharing or commenting on the information. The invention of COVID-19 vaccines has started to kick in with scientists' help. The fear of fake news can undermine people's confidence in vaccines, so the urge to identify fake news automatically can help massively to combat this virus. With the help of this project, tweets regarding COVID-19 can be verified and examined to check their credibility. This research is essential for my understanding of the fake news relevant to COVID-19. This further emphasises the aim of this whole thesis project, which is to develop an independent system that assists in recognising fake news on Twitter. The disadvantages of spreading fake news have a tremendous emotional and physical impact on elderly people who tend to believe news and theories promptly without any cross-verification. This can become life-threatening for a particular age group unwilling to check the veracity of the news. Figure 1.1 shows a tweet portraying fake information that can mislead many people and can be manipulated by many individuals. Thus, detecting fake news with automatic methods on Twitter related to COVID-19 is of vital importance.

Figure 1.1: A tweet portraying false information that can mislead and manipulate many individuals who believe this statement is true [20].

1.4 Methodology

1.4.1 Research Question

This project studies the value of both text and a social network structure for an automated misinformation detection system. The main research question to explore in this thesis – is misinformation better detected using text, or can the social network graph structure itself be a better detector?

1.4.2 Research Hypothesis

In this research, the central hypothesis to test is whether an input of a node feature representation in a graph to a supervised classifier algorithm would yield better performance than text-based features in detecting fake tweets. To the best of my knowledge, this approach fills a gap in the literature.

1.4.3 Research Methodology

In this project, both traditional NLP techniques and social network analysis are applied and compared. The project consists of two main parts. The first part is the text-based classification technique, which involves building supervised classification algorithms. These classification models classify tweets as either true/fake based on tweet text content. This research will use various text features, such as BoW, TF-IDF, and Word2Vec, and compare their performances with different classification models.

The second part uses the graph-structure classification technique, where the data is represented in a graph containing nodes and edges (connections between the nodes). Then, this graph representation will be fed as input to supervised ML algorithms. This graph representation learning approach shows the relationships between the tweet contents and the key players in promoting the information, as well as identifying the reasons that promote the information. Therefore, data will be represented in a graph in which nodes represent tweets id and edges represent the relationships between them (retweets/replies). This will allow classifying the tweet as either true/fake based on graph structure and not the tweet text.

Moreover, the node classification technique will be used to classify the tweet in the output layer. Still, it needs further processing to get it working correctly.

1.5 Objectives

A list of objectives was provided to design an automated COVID-19 misinformation detection system that identifies tweets that include fake content. These objectives aim to create such a system that assists users in verifying the truthfulness of any COVID-19 related tweets. The main objectives of this research project are as follows:

2.2.3 Word2Vec

Word2Vec is a yet another popular method of representing the document vocabulary in the form of feature vectors. It is computationally expensive since it requires training a shallow neural network architecture, but the performance gain provided by this method is very significant since it can identify which words are semantically similar and capture the context of every token in the document vocabulary. To train a neural network to learn word features, the features are initially represented in one-hot encoding. In this format, each word is represented by a vector of length equal to the size of the vocabulary. The vector has zero at all indices and one at the index corresponding to the word it is supposed to represent. In this technique, the shallow network is trained for a language-based problem which is not necessarily relevant to the problem we are attempting to solve such as auto completion of text. Once the model is trained for such a problem, it automatically learns the semantic relationship between the words. Consequently, the weights of this model can be used as the representations of tokens. The Word2Vec method has two variations [27]. The variations and their details are mentioned as follows:

- **Common Bag of Words:** In the CBoW method, the context of a given target word is used to predict that word. Context of a word refers to the surrounding n-grams of a word. The input layer has a dimension equal to kxV where k is the number of context words takes and V is the number of unique words in the vocabulary. The output layer has a dimension equal to V. The one-hot representations of each word are fed into a separate input layer. These representations are then multiplied with corresponding weight matrices and fed into the hidden layer. The weight matrix corresponding to the output of the hidden layer serves as the word representation which is extracted after training the shallow neural network.

- **Skip-Gram model:** The input and output dimensions of the skip-gram model is opposite to the CBoW model. In this case, we use a specific word to predict the context. The input layer has a dimension equal to V, where V is a specific word. The output layer has a dimension of kxV where k is the number of tokens in the context and V is the size of vocabulary. In case of this model, the weight matrix between the input layer and the hidden layer is used as the representation for each word after training the model.

2.3 Graph Terminology and Representation

This section illustrates some basic knowledge about data graph representation.

2.3.1 Graph Representation

Graph representation comprises data that are connected to each other. The critical point to highlight is that graph data is similar to any other data type. However, the main difference between graph data and other data types is that graph data is structured data. This means that in addition to defining the data points, the relation between the data points should be defined as well, which provides us with a structure. This extra information is often valuable in analyzing the data, so graphs and graph neural networks represent better performance than other data structures.

2.3.2 Nodes and Edges

Nodes are the data points that are represented in the graph. What makes graph data different from other types of data is the representation of edges that connect to nodes or data points that are similar. Hence, if two nodes are related or identical, there will be an edge connecting the nodes. In this context, the nodes could present the tweet text, and the edges are the communication between Twitter users by retweeting/replying to a given tweet.

2.3.3 Features

The recent papers in the neural network domain [37, 24] allow us to introduce a piece of additional information in the form of features. These features are the extra information that each node could carry on. This information in the form of a feature could be the number of followers, likes, total replies, etc.

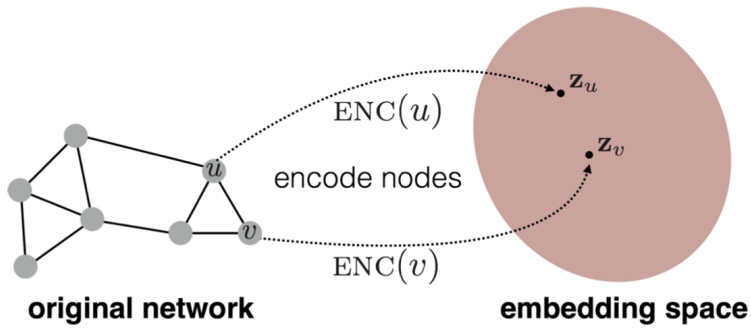

Figure 2.1: A figure illustrating node embedding learning methodology where we have the encoder (ENC) that maps nodes to a low-dimensional embedding space in a way that show the relative positions of the nodes in the original graph. [17] .

2.3.4 Directed Graph

In the case of a directed graph, an arrow shows how the nodes are related in this case. For example, node A is connected to node B, but node B is not connected to node A. So what we do is we draw an arrow from the source (Node A) to the target (Node B). Twitter is an example of a directed graph in the social media network where if the first person follows the second person, the second person does not need to follow back or follow the first person. To mathematically represent graph data, this can be shown in the following notation:

$$G = (V, E, u) \tag{2.10}$$

where G is the graph, V describes vertices (nodes), and E is a vector that describes edges or links, including two types of the matrices adjacency matrix and weights matrix. u represents the feature vector.

2.3.5 Undirected Graph

In an undirected graph, there are two arrows or no arrows at all to define an undirected graph. Moreover, in an undirected graph, the relation between the nodes is symmetric. This implies that when node A is connected to Node C, Node C is also connected to node A.

2.4 Representation Learning on Graphs

This part illustrates the approach behind learning node embeddings. Likewise, demonstrates the Node2Vec embedding algorithm which is used in the project implementation described in chapter 3

2.4.1 Learning Node Embeddings

One of the primary challenges faced in graph representation learning is to select an embedding method for the graph nodes [17]. The aim of learning the appropriate feature representation of nodes is to transform these nodes to an embedding space and use the resultant transformation of each node as a low-dimensional vector as an input to the classification algorithm. One of the most used approaches for learning node embeddings is the use of an encoder-decoder model. Figure 2.1 shows an overview of node embedding learning approach. The encoder part is responsible for mapping the nodes into the embedding space using a well devised methodology. Usually, a shallow embedding approach is employed where embedding function is simply a lookup function which extracts the embedding vector from the embedding matrix when given a unique node id using the strategy that had been devised to generate the embedding matrix. On the other hand, the decoder part of the network is used to map the feature representations or embeddings to the respective node or structural and positional information about that node such that the node and its neighborhood can be reconstructed using this information.

The pseudocode for creating the dataset can be seen in Algorithm 3.1. The whole procedure was carried out using json and pandas data-frame. By the end, the CSV file is stored in an empty list created initially and that is to be used further.

3.4 Dataset Overview

The main dataset used in the project implementation contains 237743 records and 28 features. The dataset comprised of nine columns, where first column corresponded the personality type and second column corresponded to the post. Figure 3.3 shows an example of the 10 rows of the dataset. The first column represents the tweet ID, the second column shows the Id of the author who posted the tweets on Twitter. Likewise, the third colum illustrates the device that was used to post twitter posting. Similarly, the fourth column shows the text of the tweet. The fifth column contains one of two labels (i) True or (ii) Fake. The true tweets are annotated with 1, and fake tweets are annotated as 0. The dataset contains 179681 tweets while 58062 tweets are labeled as Fake. The six column contains the number of retweets of a particular tweet. In addition, the seven column represents the number of replies of the tweets while column eights reveals the number of likes. Finally, the last column shows the counts of quotes.

For example, in Figure 3.3 second row, the number 4056 in the six column represents the number of retweets of that tweet. Similarly, the column seven, eight, nine, and 10 represents public reaction in the form of reply, like, and quote. Finally, on average each tweets in both labels (True and Fake) contains approximately 150 words.

	id	author_id	source	text	label	retweet	reply	like	quote
0	1357411153000000000	1289288731000000000	Twitter for Android	Should there even be an All-Star Game this yea...	True	0	0	0	0
1	1357411153000000000	1275531383000000000	Twitter for Android	RT @narendramodi: Spoke to President @CyrilRam...	True	4056	0	0	0
2	1357411152000000000	4825557477	Twitter for iPhone	RT @TorontoStar: How risky are Ontario schools...	True	43	0	0	0
3	1357411152000000000	2866489102	Twitter Web App	RT @dankennedy_nu: Now up at @storybench: How ...	True	1	0	0	0
4	1357411152000000000	22970986	SocialNewsDesk	'Don't treat this lightly': Coronavirus surviv...	True	0	0	0	0
5	1357411152000000000	1211482699000000000	Twitter for iPhone	This is considered a felony with Covid going o...	True	0	0	0	0
6	1357411151000000000	1891191457	Twitter Web App	@SteveLawrence_ You would say that but every v...	True	0	0	0	0
7	1357411151000000000	1293819643000000000	Twitter for iPhone	RT @tiwtter2345: #闫丽梦 \n#LimengYan \n\n#Covid1...	True	2	0	0	0
8	1357411151000000000	2578851859	Twitter for iPhone	RT @HelenaMorenoLA: Last year we didn't know C...	True	1	0	0	0
9	1357411151000000000	103154813	Twitter for iPhone	RT @NoEmmeG: man we could roll out so many suc...	True	23	0	0	0

Figure 3.3: The first 10 rows of the dataset

3.4.1 Data Wrangling

Here I explain the steps taken to prepare the given large scale data into a suitable form for the study analysis. Overall, I read the JSON data files into a large pandas dataframe and then export it into a CSV file. Figure 3.4 shows a sample of the data in JSON format.

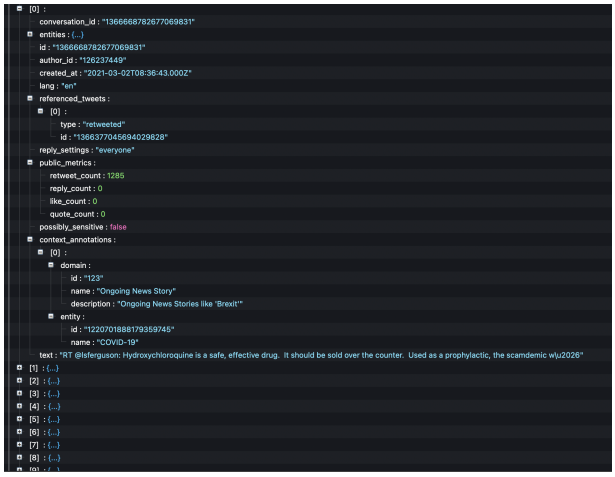

Figure 3.4: Nested JSON Dataset Structure

3. Pre-processing of data to be in the right form to function correctly when feeding it to a classification model.

4. Representing text data using BoW text feature.

5. Representing text data using TF-IDF text feature.

6. Representing text data using Word2Vec text feature.

7. Representing the data in a graph and using Node2Vec algorithm to automatically learn the feature representation.

8. Development of the prediction model.

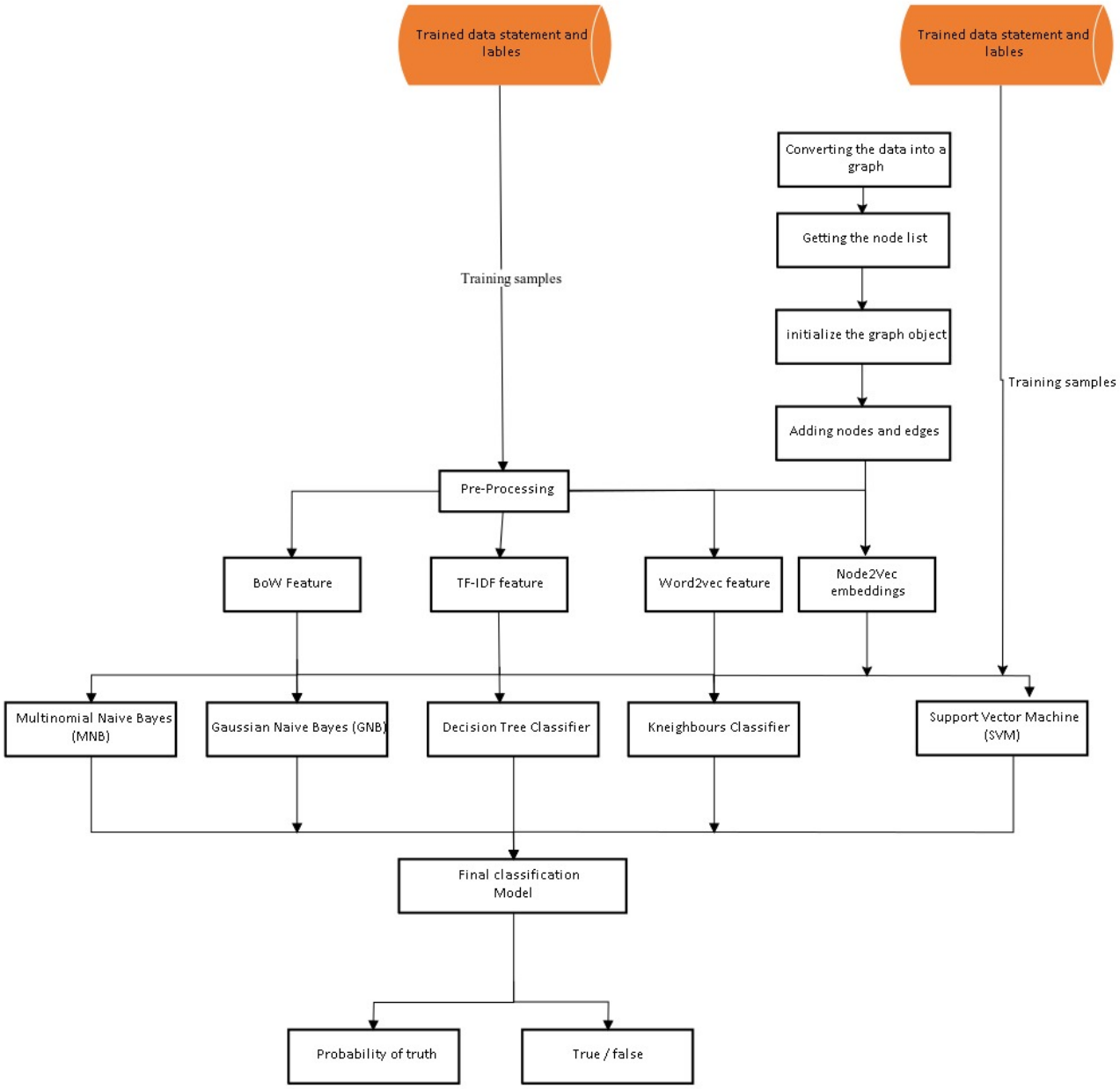

Figure 3.9: The architecture of the proposed pipeline.

3.5.1 Data Pre-processing

Textual data in social media is unstructured; various text analytics have been performed to represent it in a structured fashion. Information retrieval, document classification, topic modelling, and natural language processing are all part of text mining (NLP). NLP enables computers to effectively grasp and analyse human language. NLP's fundamental goal is to extract information from a large text corpus. Pattern matching, normalization, lemmatization, tokenization, tagging, linguistic and lexical analysis are all tasks handled by text pre-processing in natural language processing pipelines. [19]. In this study I used a number of steps to pre-process the data:

- Cleaning up the text contents by lowering case the tweet text.
- Removing retweets tag "RT" from the tweet text as it does not add value to the meaning of the text.
- Removing link website "http" from the text.
- Removing mention symbol (@) from the text.
- Removing hashtag sign from the text.
- Removing punctuation from the text.
- Removing emojies from the text.
- Deleting short words such as be, a, as..etc
- Removing stop words from the tweet text.
- Text tokenisation

3.5.2 Feature Extraction

I conducted experiments to build classification models relying only on one feature which is BoW before focusing on extracting more features to improve the performance. In this study, I chose the following features to experiemnt with:

1. BoW
2. TF-IDF
3. Word2Vec
4. Node2Vec for feature learning on graphs.

For the first part of the study, I simply look at the tweet text to classify tweets. For the second part of the study, I study the effect of graph structure classification where I check the retweets of a given tweet to investigate the dissemination of disinformation. The namely incorrect and partially false tweets and compute the pace of retweets as an agent for the speed of propagation. A retweet is a tweet re-posting, which may be done by or without a remark by a Twitter user [14].

Classifier	Features
MNB	BoW TF-IDF Word2Vec
GNB	BoW TF-IDF Word2Vec
KNN	BoW TF-IDF Word2Vec
DT	BoW TF-IDF Word2Vec

Table 3.1: The classifiers with various independent feature representation

4. Taking the experiment a step ahead to try different feature combination with each classifier as the following:

   - **BoW and TF-IDF**
   - **BoW and Word2Vec**
   - **TF-IDF and Word2Vec**
   - **BoW, TF-IDF and Word2Vec**

3.7 Graph Modelling

This part of the research involves representing the data in a graph structure and getting the embeddings of the structure using Node2Vec algorithm. The graph consists of tweet id as source nodes and retweets/replies as connections between those nodes (destination nodes). This will enable us to see how many retweets/replies each tweet receive. Hence, classifying the tweet based on its structure not the text itself. Figure 3.11 shows the built graph.

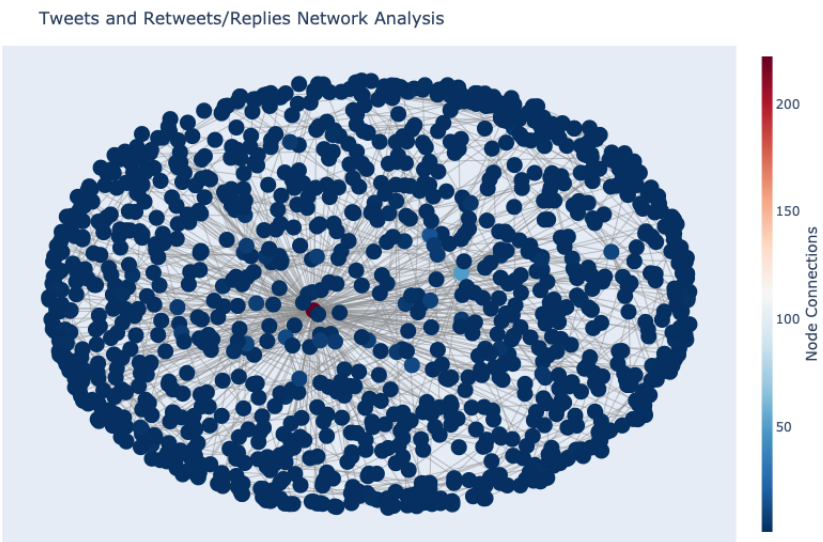

Figure 3.11: A graph representing the tweet ID and its retweets/replies connections

Classifiers	Fake Class			True Class			ROC	Acc	Micro	Weighted
	Prec	Recall	F1-Fake	Prec	Recall	F1-True			avg F-1	avg F-1
MNB	**0.64**	0.17	0.27	0.78	**0.97**	**0.87**	**0.69**	**0.77**	**0.57**	**0.72**
GNB	0.32	**0.42**	**0.36**	**0.79**	0.71	0.75	0.55	0.63	0.55	0.65
DT	0.19	0.07	0.11	0.75	0.90	0.82	0.46	0.69	0.46	0.64
KNN	0.18	0.05	0.08	0.75	0.92	0.83	0.55	0.70	0.45	0.65

Table 4.4: TF-IDF based feature with various classifiers

Regarding Word2Vec-based features, if we evaluate the results with having an accuracy of 0.75 for MNB, 0.80 for GNB, 0.59 for DT, and 0.73 for KNN as shown in Table 4.5. We also get ROC results having values of 0.80 for MNB, 0.81 for GNB, 0.59 for DT, and 0.73 for KNN. If I compare the results of classifiers and their performance by using different performance metrics, I concluded that GNB classifier has a better performance for Word2Vec features by considering both the accuracy and ROC metrics.

Classifiers	Fake Class			True Class			ROC	Acc	Micro	Weighted
	Prec	Recall	F1-Fake	Prec	Recall	F1-True			avg F-1	avg F-1
MNB	0.00	0.00	0.00	0.76	**1.00**	0.86	0.80	0.75	0.43	0.65
GNB	**0.63**	0.46	**0.53**	**0.84**	0.91	**0.88**	**0.81**	**0.80**	**0.70**	**0.79**
DT	0.30	**0.51**	0.38	0.80	0.62	0.70	0.53	0.59	0.54	0.62
KNN	0.45	0.47	0.46	0.83	0.81	0.82	0.66	0.73	0.64	0.73

Table 4.5: Word2Vec-based feature with various classifiers

These tables 4.2 4.4 4.5 show the detailed comparison of performances of the given classifiers. Lastly, if we compare three features along with both classes as true and fake and want to compute the results as a whole on given classifiers, we can see that from the results classifiers perform better with Word2Vec based features. This is because all of the performance metrics have high throughput showing that classifiers have a better prediction of classes with having Word2Vec features. This is because Word2Vec features vectorises the text objects for better classification and prediction purposes. Moreover, Word2Vec was the most accurate out of all independent features as shown in Figure 4.1

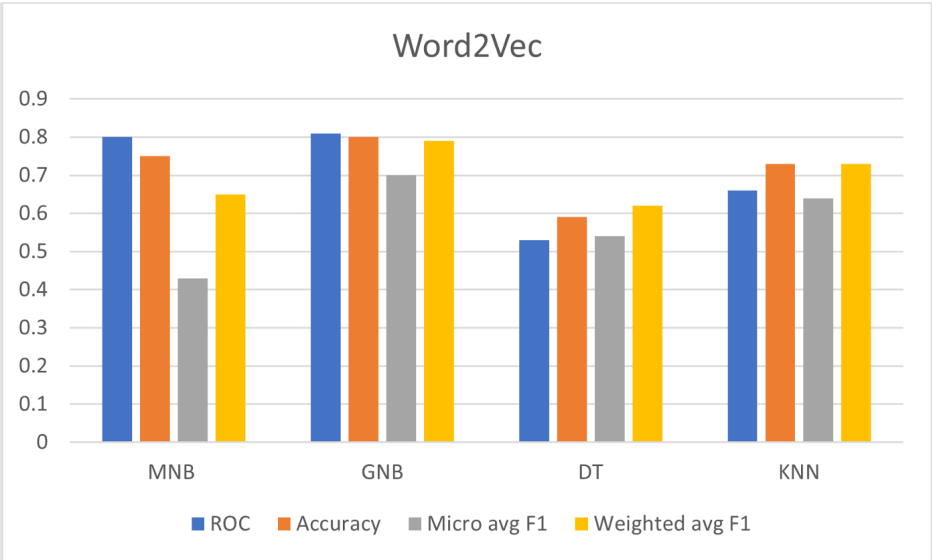

Figure 4.1: Usage of Word2Vec features generates the most accurate results

4.6 ML Models with Combined Features Results

Here I have decided to move from single features to a combination of features. I run the classifiers on the combination of features like first, I decided to combine BoW and TF-IDF. Then, BoW was combined with Word2Vec, TF-IDF and Word2Vec was combined, lastly, BoW was combined with TF-IDF and Word2Vec. After the combination, five different classifiers were selected to evaluate the results for classification. I again here classify the dataset into two classes as true and fake. The performance of these methods is compared and evaluated using multiple metrics including precision, recall, F1-score, and ROC measures. After measuring performance metrics we then calculate the accuracy of the classifiers for accurately classifying text data into its respective class.

The first features were combined and computed results using SVM classifier as shown in 4.6. The accuracy of the classifier in classification is 0.83 and has an ROC of 0.83 for the (BoW + TF-IDF) combination. Again, SVM was run with a combination of (BoW + Word2Vec) and got an accuracy of 0.83. From the above-generated results by the classifiers, we can conclude that SVM performs better with features combination.

Again, all of the feature combinations are given to the classifiers as shown in Tables 4.6, 4.7, 4.8, 4.9 and 4.10 for evaluating the results and make a comparison between all the classifiers and show which of the classier has a high true positive rate. If we compare all the results, we can conclude with the fact that despite the combination of the feature we still have the classifier SVM which can perform better in this case as well. As seen from the above-mentioned tables we can evaluate the results and make a statement that the SVM classifier has high accuracy which means that SVM classifies the test dataset accurately having a high true positive rate.

Classifiers	**Fake Class**			**True Class**			**ROC**	**Acc**	**Micro**	**Weighted**
	Prec	**Recall**	**F1-Fake**	**Prec**	**Recall**	**F1-True**			**avg F-1**	**avg F-1**
BoW + TF-IDF + SVM	0.75	0.52	0.61	0.86	0.94	0.90	0.83	0.83	0.76	0.83
BoW + Word2Vec + SVM	0.75	0.52	0.61	0.86	0.94	0.90	0.83	0.83	0.76	0.83

Table 4.6: Combined features with SVM classifier

Classifiers	**Fake Class**			**True Class**			**ROC**	**Acc**	**Micro**	**Weighted**
	Prec	**Recall**	**F1-Fake**	**Prec**	**Recall**	**F1-True**			**avg F-1**	**avg F-1**
BoW + TF-IDF + MNB	0.00	0.00	0.00	0.76	1.00	0.86	**0.86**	0.76	0.43	0.65
BoW + Word2Vec +MNB	0.00	0.00	0.00	0.76	1.00	0.86	0.68	0.76	0.43	0.65
TF-IDF + Word2Vec +MNB	0.00	0.00	0.00	0.76	1.00	0.86	0.75	0.76	0.43	0.65
BoW + TF-IDF + Word2Vec + MNB	0.00	0.00	0.00	0.76	1.00	0.86	0.80	0.76	0.43	0.65

Table 4.7: Combined features with MNB classifier

Classifiers	Fake Class			True Class			ROC	Acc	Micro	Weighted
	Prec	Recall	F1-Fake	Prec	Recall	F1-True			avg F-1	avg F-1
BoW + TF-IDF + GNB	0.69	**0.43**	**0.53**	**0.84**	0.94	0.88	**0.86**	0.81	**0.71**	**0.80**
BoW + Word2Vec + GNB	**0.78**	0.33	0.47	0.82	**0.97**	**0.89**	0.80	0.81	0.68	0.79
TF-IDF + Word2Vec + GNB	0.76	0.34	0.47	0.82	**0.97**	**0.89**	0.82	0.81	0.68	0.78
BoW + TF-IDF + Word2Vec + GNB	0.71	0.38	**0.49**	0.83	0.95	0.88	0.81	0.81	0.69	0.79

Table 4.8: Combined features with GNB classifier

Classifiers	Fake Class			True Class			ROC	Acc	Micro	Weighted
	Prec	Recall	F1-Fake	Prec	Recall	F1-True			avg F-1	avg F-1
BoW + TF-IDF + DT	**0.55**	**0.66**	**0.60**	**0.88**	0.83	**0.85**	**0.76**	**0.78**	**0.73**	**0.79**
BoW + Word2Vec + DT	0.29	0.12	0.18	0.76	**0.90**	0.83	0.51	0.71	0.50	0.67
TF-IDF + Word2Vec + DT	0.38	0.20	0.26	0.78	**0.90**	0.83	0.54	0.72	0.55	0.69
BoW + TF-idf + Word2Vec + DT	0.46	0.32	0.38	0.80	0.88	0.84	0.59	0.74	0.61	0.72

Table 4.9: Combined features with DT classifier

values are compared to those obtained using a combination of features , Table 4.13, the highest values of ROC score, accuracy, micro average f1 score and weighted average f1 scores are 0.82, 0.81, 0.69 and 0.79 respectively. These results are 12.33%, 6.58%, 15% and 11.27% higher than those achieved by using independent features allowing us to concretely conclude that using a combination of text-based features improves classification results of machine learning models significantly.

Another observation made through the comparison of independent and combined text features was that out of all machine learning models used, MNB performs the best with the use of independent features whereas GNB works well with a combination of features. This helps improve our results further and hence it can for a baseline to be used in further research's in the future.

	ROC	**Accuracy**	**Micro avg F1**	**Weighted avg F1**
MNB	**0.73**	**0.76**	0.55	**0.71**
GNB	0.66	0.68	**0.6**	0.69
DT	0.47	0.58	0.49	0.63
KNN	0.56	0.71	0.51	0.68

Table 4.12: Average performance for all classifiers using independent features

	ROC	**Accuracy**	**Micro avg F1**	**Weighted avg F1**
MNB	0.77	0.76	0.43	0.65
GNB	**0.82**	**0.81**	**0.69**	**0.79**
DT	0.60	0.74	0.60	0.72
KNN	0.65	0.75	0.57	0.72

Table 4.13: Average performance for all classifiers using a combination of features

4.10.2 Comparison of text and graph modelling

An important purpose of this research is to determine which method is more accurate, text-based modelling or graph-based. Since many experiments were performed in the text-based part, it was decided to use combination of features in GNB classifier for comparison since it had the finest performance. The average of all values for each evaluation metrics was taken and a separate table was constructed. The comparison can be seen in Figure 4.2.

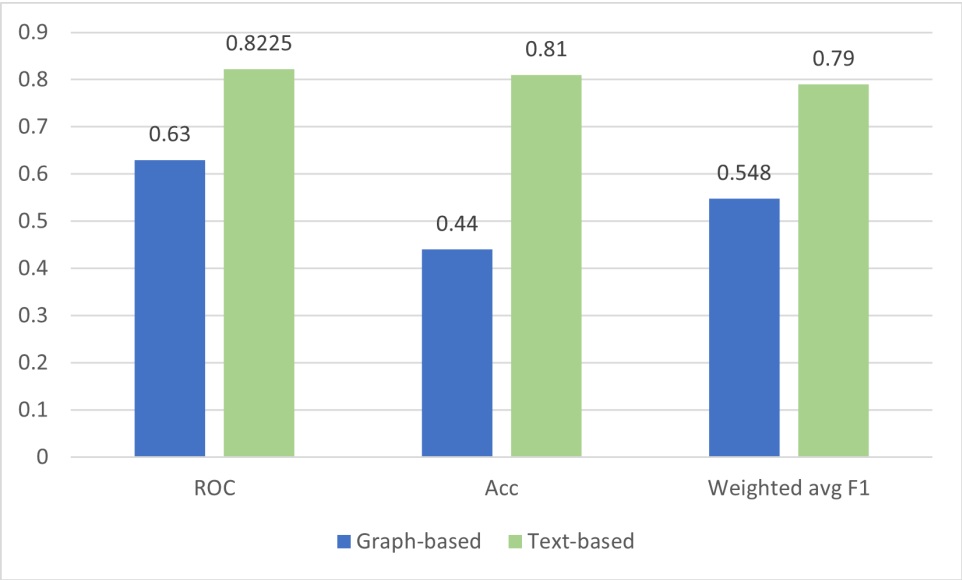

Figure 4.2: Comparison of graph-based and text-based modelling

In the figure above, it can be clearly seen that text-based modelling outperforms graph-based modelling in terms of ROC score, accuracy as well as weighted average F1 score. In the graph-based experiment,

Chapter 5

Conclusion

The current pandemic has resulted in an acceleration of the usage of technology, on a government level as well as on a personal one. Information is being diffused at an uncontrollable speed and whereas majority of it is extremely useful as well as necessary, some of it is fabricated, resulting in life-threatening outcomes. In this research, I have explored the problem of COVID-19 misinformation during the pandemic by building an automatic COVID 19 misinformation detection system. A comprehensive comparison is presented between different ML classifiers using various input features for text based and graph structure based architectures.

In this research, a combination of 7 datasets, was initially cleaned, labelled and separated from JSON data files into CSV files to be used for each experiment. After the process of data annotation and pre-processing, the results in this research were generated for 3 scenarios and subsequently compared. The scenarios were text based classification models with independent features, text based classification models with combined features and graph structure classification.

During the evaluation of usage of independent features, it was observed that MNB performed the best for BoW and TF-IDF based features in terms of ROC, accuracy and F1-score. When Word2Vec based feature was used, GNB outperformed the other classifiers by obtaining a higher accuracy, F1 score and ROC score.

Secondly, when features were combined, SVM outperformed the other classifiers in all evaluation metrics when given combinations of BoW with TF-IDF and BoW with Word2Vec.

5.1 Summary

This thesis can be summarized in the following manner:

Chapter 1 provides an introduction to the non-technical aspects of this thesis. It explains the importance of this research as well as the motivation behind it whilst also discussing the various challenges linked with the project.

Chapter 2 contains a detailed explanation of all technical aspects which need to be understood for this research. A sufficient amount of literature review is provided for various machine learning techniques used for textual data and necessary terminologies linked with this research are explained. This chapter also includes details regarding the dataset to be used for our experiments.

Chapter 3 provides detailed descriptions of the experiments performed in this research. All steps performed to achieve the output are listed in a systematic manner and are explained conspicuously. This chapter also includes explanations of the machine learning algorithms to be used and lastly all evaluation metrics used are discussed in-depth.

Chapter 4 is an extremely crucial chapter since it contains all results obtained in this research and a thorough comparison of text-based data and graph-based data using 5 different machine learning algorithms is provided as well. The results are analyzed and any shortcomings are mentioned.

5.2 Project Status

The initial objectives of this project were to:

- Labelling the dataset, adding binary labels to the dataset as true/fake is required. (Complete)
- Building a baseline model that classify each tweet and improving its performance with coming experiments. (Complete)
- Encoding BoW, TF-IDF and Word2Vec feature representation from the tweet text. (Complete)
- Training classifiers with independent feature representation. (Complete except SVM classifier with independent features)
- Training more than one feature to represent the text. (Complete)
- Represent the tweet text related information in a graph structure. (Complete)
- Get input vectors from the graph. (Complete)
- Training classifiers with graph feature representation. (Complete)
- Compare and analyse the performance of both text and graph approaches. (Complete)
- Applying graph feature representation to a GNN algorithm. (Incomplete)

Some obstacles I faced during the execution of the research implementation include the following:

- Running the following experiments:
  - SVM algorithm with independent features including BoW, TF-IDF and Word2Vec : This could not be achieved because the model took more than 48 hours to train on local machine with great difficulty.
  - Combination of BoW, TF-IDF, Word2Vec based features: This also proved to be problematic as training took a long time.

5.3 Further Work

This research provides a foundation for machine learning based approaches to handle incorrect information during COVID-19. It can be extended in multiple directions, enhancing its practicality in real-world problems.

Firstly, the graph based aspect of this research can be extended by using different graph structure types and analyzing the results. It is highly possible that other types may improve the results further.

Another direction can be to alter the labelling process of the data. In this research, data has been annotated automatically using 4 hashtag keywords. In the future, this process can be performed manually to achieve more accurate results

Lastly, subsequent to machine learning algorithms, deep learning techniques have been increasingly used. These techniques have been providing much more accurate results and have the ability to self-learn, executing complex tasks efficiently. For this specific problem, deep neural networks can be utilized to improve the results substantially in a shorter amount of time. Specifically, Node2Vec embeddings can be performed using graph neural networks since they are highly capable of understanding complex patterns in data.

Bibliography

[1] Worldometers coronavirus cases. 2021.

[2] Wasim Ahmed, Josep Vidal-Alaball, Joseph Downing, Francesc López Seguí, et al. Covid-19 and the 5g conspiracy theory: social network analysis of twitter data. *Journal of medical internet research*, 22(5):e19458, 2020.

[3] Bilal Al-Ahmad, Ala'M Al-Zoubi, Ruba Abu Khurma, and Ibrahim Aljarah. An evolutionary fake news detection method for covid-19 pandemic information. *Symmetry*, 13(6):1091, 2021.

[4] Md Al-Zaman et al. Covid-19-related social media fake news in india. *Journalism and Media*, 2(1):100–114, 2021.

[5] Vimala Balakrishnan and Ethel Lloyd-Yemoh. Stemming and lemmatization: a comparison of retrieval performances. 2014.

[6] J Scott Brennen, Felix Simon, Philip N Howard, and Rasmus Kleis Nielsen. Types, sources, and claims of covid-19 misinformation. *Reuters Institute*, 7:3–1, 2020.

[7] Limeng Cui and Dongwon Lee. Coaid: Covid-19 healthcare misinformation dataset. *arXiv preprint arXiv:2006.00885*, 2020.

[8] Padraig Cunningham and Sarah Jane Delany. k-nearest neighbour classifiers: (with python examples). *arXiv preprint arXiv:2004.04523*, 2020.

[9] CIPRIAN-GABRIEL Cușmaliuc, LUCIA-GEORGIANA Coca, and ADRIAN Iftene. Identifying fake news on twitter using naive bayes, svm and random forest distributed algorithms. In *Proceedings of The 13th Edition of the International Conference on Linguistic Resources and Tools for Processing Romanian Language (ConsILR-2018) pp*, pages 177–188, 2018.

[10] Karen M Douglas. Covid-19 conspiracy theories. *Group Processes & Intergroup Relations*, 24(2):270–275, 2021.

[11] K SRIVASTAVA Durgesh and B Lekha. Data classification using support vector machine. *Journal of theoretical and applied information technology*, 12(1):1–7, 2010.

[12] Theodoros Evgeniou and Massimiliano Pontil. Support vector machines: Theory and applications. In *Advanced Course on Artificial Intelligence*, pages 249–257. Springer, 1999.

[13] Aditya Grover and Jure Leskovec. node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 855–864, 2016.

[14] Stefano Guarino, Noemi Trino, Alessandro Chessa, and Gianni Riotta. Beyond fact-checking: Network analysis tools for monitoring disinformation in social media. In *International Conference on Complex Networks and Their Applications*, pages 436–447. Springer, 2019.

[15] Kriti Gupta. Fake news analysis and graph classification on a covid-19 twitter dataset. 2021.

[16] Abdullah Hamid, Nasrullah Shiekh, Naina Said, Kashif Ahmad, Asma Gul, Laiq Hassan, and Ala Al-Fuqaha. Fake news detection in social media using graph neural networks and nlp techniques: A covid-19 use-case. *arXiv preprint arXiv:2012.07517*, 2020.

[17] William L Hamilton. Graph representation learning. *Synthesis Lectures on Artifical Intelligence and Machine Learning*, 14(3):1–159, 2020.

[18] Yi Han, Shanika Karunasekera, and Christopher Leckie. Graph neural networks with continual learning for fake news detection from social media. *arXiv preprint arXiv:2007.03316*, 2020.

[19] Nitin Hardeniya, Jacob Perkins, Deepti Chopra, Nisheeth Joshi, and Iti Mathur. *Natural language processing: python and NLTK*. Packt Publishing Ltd, 2016.

[20] Rebecca Heilweil. Twitter now labels misleading coronavirus tweets with a misleading label, May 2020.

[21] Rajni Jindal, Ruchika Malhotra, and Abha Jain. Techniques for text classification: Literature review and current trends. *webology*, 12(2), 2015.

[22] Devyani Keskar, Sushila Palwe, and Ayushi Gupta. Fake news classification on twitter using flume, n-gram analysis, and decision tree machine learning technique. In *Proceeding of International Conference on Computational Science and Applications*, pages 139–147. Springer, 2020.

[23] Hyunuk Kim and Dylan Walker. Leveraging volunteer fact checking to identify misinformation about covid-19 in social media. *Harvard Kennedy School Misinformation Review*, 1(3), 2020.

[24] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016.

[25] Jan Kirchner and Christian Reuter. Countering fake news: A comparison of possible solutions regarding user acceptance and effectiveness. *Proceedings of the ACM on Human-computer Interaction*, 4(CSCW2):1–27, 2020.

[26] Ramez Kouzy, Joseph Abi Jaoude, Afif Kraitem, Molly B El Alam, Basil Karam, Elio Adib, Jabra Zarka, Cindy Traboulsi, Elie W Akl, and Khalil Baddour. Coronavirus goes viral: quantifying the covid-19 misinformation epidemic on twitter. *Cureus*, 12(3), 2020.

[27] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013.

[28] Azzam Mourad, Ali Srour, Haidar Harmanai, Cathia Jenainati, and Mohamad Arafeh. Critical impact of social networks infodemic on defeating coronavirus covid-19 pandemic: Twitter-based study and research directions. *IEEE Transactions on Network and Service Management*, 17(4):2145–2155, 2020.

[29] Arjun Mukherjee, Vivek Venkataraman, Bing Liu, and Natalie Glance. What yelp fake review filter might be doing? In *Seventh international AAAI conference on weblogs and social media*, 2013.

[30] Harsh H Patel and Purvi Prajapati. Study and analysis of decision tree based classification algorithms. *International Journal of Computer Sciences and Engineering*, 6(10):74–78, 2018.

[31] Parth Patwa, Shivam Sharma, Srinivas Pykl, Vineeth Guptha, Gitanjali Kumari, Md Shad Akhtar, Asif Ekbal, Amitava Das, and Tanmoy Chakraborty. Fighting an infodemic: Covid-19 fake news dataset. In *International Workshop on Combating On line Ho st ile Posts in Regional Languages dur ing Emerge ncy Si tuation*, pages 21–29. Springer, 2021.

[32] Juan Pizarro. Using n-grams to detect fake news spreaders on twitter. In *CLEF (Working Notes)*, 2020.

[33] Adi Prasetyo, Bayu Dwi Septianto, Guruh Fajar Shidik, and Ahmad Zainul Fanani. Evaluation of feature extraction tf-idf in indonesian hoax news classification. In *2019 International Seminar on Application for Technology of Information and Communication (iSemantic)*, pages 1–6. IEEE, 2019.

[34] Lotte Pummerer, Robert Böhm, Lau Lilleholt, Kevin Winter, Ingo Zettler, and Kai Sassenberg. Conspiracy theories and their societal effects during the covid-19 pandemic. *Social Psychological and Personality Science*, page 19485506211000217, 2020.

[35] Shahzad Qaiser and Ramsha Ali. Text mining: use of tf-idf to examine the relevance of words to documents. *International Journal of Computer Applications*, 181(1):25–29, 2018.

[36] Irina Rish et al. An empirical study of the naive bayes classifier. In *IJCAI 2001 workshop on empirical methods in artificial intelligence*, volume 3, pages 41–46, 2001.

[37] Franco Scarselli, Marco Gori, Ah Chung Tsoi, Markus Hagenbuchner, and Gabriele Monfardini. The graph neural network model. *IEEE transactions on neural networks*, 20(1):61–80, 2008.

[38] Yunita Dwi Setiyaningrum, Annisa Fitrianingtiyas Herdajanti, Catur Supriyanto, et al. Classification of twitter contents using chi-square and k-nearest neighbour algorithm. In *2019 International Seminar on Application for Technology of Information and Communication (iSemantic)*, pages 1–4. IEEE, 2019.

[39] Gautam Kishore Shahi, Anne Dirkson, and Tim A Majchrzak. An exploratory study of covid-19 misinformation on twitter. *Online social networks and media*, 22:100104, 2021.

[40] Manoj Kumar Sharma, Nitin Anand, Akash Vishwakarma, Maya Sahu, Pranjali Chakraborty Thakur, Ishita Mondal, Priya Singh, Ajith Sj, et al. Mental health issues mediate social media use in rumors: implication for media based mental health literacy. *Asian journal of psychiatry*, 53:102132, 2020.

[41] Mandeep Singh, Mohammed Wasim Bhatt, Harpreet Singh Bedi, and Umang Mishra. Performance of bernoulli's naive bayes classifier in the detection of fake news. *Materials Today: Proceedings*, 2020.

[42] Yan-Yan Song and LU Ying. Decision tree methods: applications for classification and prediction. *Shanghai archives of psychiatry*, 27(2):130, 2015.

[43] Sander van Der Linden, Jon Roozenbeek, and Josh Compton. Inoculating against fake news about covid-19. *Frontiers in psychology*, 11:2928, 2020.

[44] Youze Wang, Shengsheng Qian, Jun Hu, Quan Fang, and Changsheng Xu. Fake news detection via knowledge-driven multimodal graph convolutional networks. In *Proceedings of the 2020 International Conference on Multimedia Retrieval*, pages 540–547, 2020.

[45] Xichen Zhang and Ali A Ghorbani. An overview of online fake news: Characterization, detection, and discussion. *Information Processing & Management*, 57(2):102025, 2020.

[46] Yin Zhang, Rong Jin, and Zhi-Hua Zhou. Understanding bag-of-words model: a statistical framework. *International Journal of Machine Learning and Cybernetics*, 1(1-4):43–52, 2010.

[47] Xinyi Zhou and Reza Zafarani. A survey of fake news: Fundamental theories, detection methods, and opportunities. *ACM Computing Surveys (CSUR)*, 53(5):1–40, 2020.